

CHAPTER II.2-3, WEEK 3

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- 2.14. (a) Let S be a graded ring. Show that $\text{Proj } S = \emptyset$ if and only if every element of S_+ is nilpotent.
- (b) Let $\varphi : S \rightarrow T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \text{Proj } T : \mathfrak{p} \not\supset \varphi(S_+)\}$. Show that U is an open subset of $\text{Proj } T$, and show that φ determines a natural morphism $f : U \rightarrow \text{Proj } S$.
- (c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi : S_d \rightarrow T_d$ is an isomorphism for all $d \geq d_0$, where d_0 is any integer. Then show that $U = \text{Proj } T$ and the morphism $f : \text{Proj } T \rightarrow \text{Proj } S$ is an isomorphism.

Proof. (a) Denote $\text{nil}' S := \bigcap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over all homogeneous prime ideals of S . It suffices to show that $\text{nil}' S = \text{nil } S$.

For any prime ideal $\mathfrak{p}$, the homogenization $h(\mathfrak{p}) \subset \mathfrak{p}$ generated by its homogeneous elements is clearly a homogeneous prime ideal. Then we have

$$\text{nil}' S \supset \text{nil } S = \bigcap_{\mathfrak{p}} \mathfrak{p} \supset \bigcap_{\mathfrak{p}} h(\mathfrak{p}) = \text{nil}' S,$$

which gives us $\text{nil}' S \subset \text{nil } S$. We conclude that $\text{nil } S = \text{nil}' S$. Now, $\text{Proj } S = \emptyset$ iff $S_+ \subset \text{nil}' S = \text{nil } S$, which is exactly our result.

(b) Let $\mathfrak{p} \in \text{Proj } T$ be a homogeneous prime ideal. Then $\varphi^{-1}(\mathfrak{p})$ is a homogeneous prime ideal, and $\varphi^{-1}(\mathfrak{p}) \supset S_+$ implies

$$\mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(S_+),$$

i.e. $\mathfrak{p} \notin U$. Conversely, $\mathfrak{p} \supset \varphi(S_+)$ implies

$$\varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(S_+)) \supset S_+,$$

so that $\varphi^{-1}(\mathfrak{p}) \notin \text{Proj } S$. We conclude that U is the inverse image of $\text{Proj } S$ under the mapping of homogeneous prime ideals $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$. Hence, we have a natural map $f : U \rightarrow \text{Proj } S : \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.

Fix a homogeneous $g \in T_+$. Then

$$\varphi^{-1}(\mathfrak{p}) \ni g \implies \mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \ni \varphi(g) \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(g)) \ni g$$

gives us that

$$f^{-1}(D_+(g)) = D_+(\varphi(g)) \subset U.$$

Hence, $f : U \rightarrow \text{Proj } S$ is continuous. We must now define the map $f^\# : \mathcal{O}_S \rightarrow f_*(\mathcal{O}_T|_U)$ of sheaves.

It suffices to define $f^\#$ on basic open sets. Fix a homogeneous $g \in S_+$, and note that

$$f_*(\mathcal{O}_T|_U)(D_+(g)) = \mathcal{O}_T(f^{-1}(D_+(g)) \cap U) = \mathcal{O}_T(D_+(\varphi(g))) \cong T_{(\varphi(g))}.$$

Recall as that φ is degree preserving, so each of its localizations are as well. In particular, $\varphi_g : S_g \rightarrow T_g$ restricts to a map $\varphi_g : S_{(g)} \rightarrow T_{(\varphi(g))}$. Then the following commutative diagram gives us our sheaf map on $D_+(g)$.

$$\begin{array}{ccc} \mathcal{O}_S(D_+(g)) & \xrightarrow{f^\#(D_+(g))} & f_*(\mathcal{O}_T|_U)(D_+(g)) \\ \downarrow \wr & & \downarrow \wr \\ S_{(g)} & \xrightarrow{\varphi_g} & T_{(\varphi(g))} \end{array}$$

To show that these maps glue together, we note that $D(g) \cap D(h) = D(gh)$, and consider the following commutative diagram.

$$\begin{array}{ccc} S_{(g)} & \xrightarrow{\varphi_g} & S_{(\varphi(g))} \\ & \searrow & \searrow \\ & S_{(gh)} & \xrightarrow{\varphi_{gh}} S_{(\varphi(gh))} \\ & \nearrow & \nearrow \\ S_{(h)} & \xrightarrow{\varphi_h} & S_{(\varphi(h))} \end{array}$$

This defines our sheaf homomorphism on all of $\text{Proj } S$, so that we're done.

(c) Fix d_0 such that $\varphi : S_d \rightarrow T_d$ is an isomorphism for all $d \geq d_0$. We wish to show that $U = \text{Proj } T$, so it suffices to show that $\sqrt{\varphi(S_+)} = T_+$. One inclusion is clear since φ is degree preserving. Surjectivity of $\varphi : S_d \rightarrow T_d$ gives us $\varphi(S_+) \supset \bigoplus_{d \geq d_0} T_d$. Then, for any element $a \in T_+$, a^{d_0} is of degree at least d_0 , so $a^{d_0} \in \varphi(S_+)$ implies $a \in \sqrt{\varphi(S_+)}$. We conclude that $\sqrt{\varphi(S_+)} = T_+$, so $U = \text{Proj } T$.

It suffices to show that $f^\#$ is an isomorphism on open sets $D(g)$. Fix homogeneous $g \in S_+$. Since $D(g^n) = D(g)$ for all $n > 0$, we may assume g is of degree $\geq d_0$. We show the map $\varphi_g : S_{(g)} \rightarrow T_{(\varphi(g))}$ is surjective. For any $b/\varphi(g)^n \in (B_A)_{\mathfrak{p}}$, $b\varphi(g) = \varphi(a)$ with $a \in A$ gives us

$$\varphi_g(a/g^{n+1}) = \varphi(ag)/\varphi(g^{n+1}) = b/\varphi(g)^n \in \text{im } \varphi_g.$$

It suffices to check injectivity. If $\varphi(a/g) = 0$, then we can find $n \geq 1$ such that $\varphi(g)^n \varphi(a) = 0$. Then $\varphi(g^n a) = 0$ implies $g^n a = 0$ by injectivity of φ , whence $a/g = 0$ in $S_{(g)}$. The result follows. $\square$

2.18. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

- (a) Let A be a ring, $X = \text{Spec } A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.
- (b) Let $\varphi : A \rightarrow B$ be a homomorphism of rings, and let $f : Y = \text{Spec } B \rightarrow X = \text{Spec } A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the map of sheaves $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is injective. Show furthermore in that case f is *dominant*, i.e., $f(Y)$ is dense in X .
- (c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X , and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective.
- (d) Prove the converse to (c). Namely, if $f : Y \rightarrow X$ is a homeomorphism onto a closed subset, and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective, then φ is surjective.

Proof. (a) The result is trivial, as $D(f) = \emptyset$ iff $f \in \bigcap_{\mathfrak{p}} \mathfrak{p} = \text{nil } f$.

(b) In the language of coherent sheaves, we have $\mathcal{O}_X = \widetilde{A}$ and $f_*\mathcal{O}_Y = f_*\widetilde{B} = \widetilde{B_A}$. Then $f^\#_{\mathfrak{p}} : \mathcal{O}_{X,\mathfrak{p}} \rightarrow (f_*\mathcal{O}_Y)_{\mathfrak{p}}$ corresponds to the map $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \rightarrow (B_A)_{\mathfrak{p}}$. By CA, we have that φ is injective iff each $\varphi_{\mathfrak{p}}$ is injective, which by the above occurs iff $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is injective.

It suffices to show that $f(Y)$ is dense in X . Fix $g \in A$. Then

$$\begin{aligned} f(Y) \cap D(g) = \emptyset &\iff f^{-1}(D(g)) = \emptyset \\ &\iff D(\varphi(g)) = \emptyset \\ &\iff \varphi(g) \in \text{nil } B \\ &\iff g \in \text{nil } A \\ &\iff D(g) = \emptyset \end{aligned}$$

by (a) and injectivity of φ respectively. Any nonempty open $U \subset X$ contains a nonempty distinguished open of X , so $f(Y)$ is dense in X .

(c) First, suppose that $\varphi : A \rightarrow B$ is surjective. Localization is exact, so each of the stalks $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \rightarrow (B_A)_{\mathfrak{p}}$ is surjective, whence $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective. Surjectivity of φ gives us

$$\mathfrak{p} \supset \mathfrak{a} \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{a}) \implies \mathfrak{p} = \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(\varphi^{-1}(\mathfrak{a})) = \mathfrak{a},$$

so that

$$\begin{aligned} f(V(\mathfrak{a})) &= \{\varphi^{-1}(\mathfrak{p}) \in X : \mathfrak{p} \supset \mathfrak{a}\} \\ &= f(Y) \cap V(\varphi^{-1}(\mathfrak{a})). \end{aligned}$$

In particular,

$$f(Y) = f(Y) \cap V(\ker \varphi) \subset V(\ker \varphi).$$

Take $\mathfrak{q} \in V(\ker \varphi)$, and suppose there exists $a \in \varphi^{-1}(\varphi(\mathfrak{q})) \setminus \mathfrak{q}$. Then there is some $a' \in \mathfrak{q}$ with $\varphi(a) = \varphi(a')$, so that $a - a' \in \ker \varphi \setminus \mathfrak{q} = \emptyset$, a contradiction. Thus, $\varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$.

We now show that $\varphi(\mathfrak{q})$ is prime. Suppose $bb' \in \varphi(\mathfrak{q})$, and take $a, a' \in A$ such that $\varphi(a) = b$ and $\varphi(a') = b'$. Then $aa' \in \varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$ implies $a \in \mathfrak{q}$ or $a' \in \mathfrak{q}$. Hence, $b \in \varphi(\mathfrak{q})$ or $b' \in \varphi(\mathfrak{q})$ gives us that $\varphi(\mathfrak{q})$ is prime, so that $g : V(\ker \varphi) \rightarrow Y : \mathfrak{q} \rightarrow \varphi(\mathfrak{q})$ is a set-theoretic inverse to $f : Y \rightarrow V(\ker \varphi) \subset X$. In particular, $f(Y) = V(\ker \varphi)$.

It suffices to show that $g : V(\ker \varphi) \rightarrow Y$ is continuous. We write

$$f(V(\mathfrak{a})) = f(V) \cap V(\varphi^{-1}(\mathfrak{a})) = V(\ker \varphi + \varphi^{-1}(\mathfrak{a})),$$

so that $g^{-1}(V(\mathfrak{a})) = V(\ker \varphi + \varphi^{-1}(\mathfrak{a}))$ is closed in X . We conclude that f is a homeomorphism onto its closed image $f(Y) = V(\ker \varphi)$.

(d) Suppose now that $f : Y \rightarrow X$ is a homeomorphism onto a closed subset, and $f^\# : \mathcal{O}_X \rightarrow \mathcal{O}_Y$ is surjective. Let $Z = \text{Spec}(A/\ker \varphi)$, so that the universal property of quotients supplies $\overline{\varphi} : A/\ker \varphi \rightarrow B$ such that $\varphi = \overline{\varphi} \circ \pi$. We denote the associated morphism of sheaves $\overline{f} : Y \rightarrow Z$. Note that $\pi : A \rightarrow A/\ker \varphi$ is surjective, so (c) gives us that the associated map $g : Z \rightarrow V(\ker \varphi) \subset X$ is a homeomorphism.

$\overline{\varphi}$ is injective, so (b) implies $\overline{f}^\# : \mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is injective, and $\overline{f}(Y) = f(Y)$ is dense. $f(Y) = f(Y)^-$ is closed, so $\overline{f} : Y \rightarrow Z$ is surjective. $f = g \circ \overline{f}$ with f, g homeomorphisms onto their image implies $\overline{f}$ is a homeomorphism onto its image $\overline{f}(Y) = Z$. Additionally, $f^\# = \overline{f}^\# g^\#$ surjective implies $\overline{f}^\#$ is surjective, so that $\overline{f}^\# : \mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is an isomorphism of sheaves. Then

$$\varphi = \overline{\varphi} \circ \pi = \overline{f}^\#(Z) \circ \pi$$

is the composition of a surjective map with an isomorphism, and is therefore surjective itself. $\square$

- 3.1. Show that a morphism $f : X \rightarrow Y$ is locally of finite type if and only if *every* affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ can be covered by open affine subsets $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra.

Proof. One direction is clear. Suppose conversely that $f : X \rightarrow Y$ is locally of finite type, and fix $\operatorname{Spec} A \cong U \subset X$ and $\operatorname{Spec} B \cong V \subset Y$ affine, with $f(U) \subset V$. To proceed, we must first prove the following lemma.

Lemma. Fix a sheaf morphism $f : X \rightarrow Y$ of finite type, $\operatorname{Spec} A \cong U \subset X$, $\operatorname{Spec} B \cong V \subset Y$, $f(U) \subset V$ as above, and denote the map of rings associated to $f|_U : U \rightarrow V$ by $\varphi : B \rightarrow A$. There exist $f_{11}, \dots, f_{rs} \in A$ and $g_1, \dots, g_r \in B$ such that

$$\operatorname{Spec} A = D(f_{11}) \cup \dots \cup D(f_{rs}),$$

$$\operatorname{Spec} B = D(g_1) \cup \dots \cup D(g_r),$$

$$f(D(f_{ij})) \subset D(g_i),$$

and the induced map

$$\varphi_{g_i} : B_{g_i} \rightarrow A_{f_{ij}}$$

is of finite type.

Proof. We have an affine cover $\{V_i = \operatorname{Spec} B_i\}$ of X such that each $f^{-1}(V_i)$ can be covered by open affines $U_{ij} = \operatorname{Spec} A_{ij}$, where A_{ij} is a finitely generated B_i algebra. Fix some i and write

$$V_i \cap V = \bigcup_{D(g_\beta) \subset V_i \cap V} D(g_\beta),$$

where $D(g_\beta)$ is affine in both V_i and V . Then $f^{-1}(D(g_\beta)) = D(\varphi(g_\beta))$ and fixing some j allows us to write

$$\begin{aligned} U_{ij} \cap U &= \bigcup_{D(f_\alpha) \subset U_{ij} \cap U} D(f_\alpha) \\ &= \bigcup_{\substack{D(f_\alpha) \subset U_{ij} \cap U \\ D(g_\beta) \subset V_i \cap V}} D(f_\alpha \varphi(g_\beta)), \end{aligned}$$

where each $D(f_\alpha)$ and $D(\varphi(g_\beta))$ is distinguished in both U and U_{ij} . We denote $f_{\alpha\beta} := f_\alpha \varphi(g_\beta)$. Then $f|_{D(f_{\alpha\beta})} : D(f_{\alpha\beta}) \rightarrow D(g_\beta)$ induces the map $\varphi_{g_\beta} : B_{g_\beta} \rightarrow A_{f_{\alpha\beta}}$.

For any $g_\beta \in B$, $f_\alpha \in A$, and $f_{\alpha\beta} \in A$ as above, we write $g'_\beta \in B_i$, $f'_\alpha \in A_{ij}$, and $f'_{\alpha\beta} \in A_{ij}$ for the corresponding elements which determine the same distinguished subset. Then

$$(B_i)_{g'_\beta} = \mathcal{O}_{V_i}(D(g'_\beta)) = \mathcal{O}_V(D(g_\beta)) = B_{g_\beta},$$

$$(A_{ij})_{f'_\alpha} = \mathcal{O}_{U_{ij}}(D(f'_\alpha)) = \mathcal{O}_U(D(f_\alpha)) = A_{f_\alpha},$$

and

$$(A_{ij})_{f'_{\alpha\beta}} = A_{f_{\alpha\beta}} = A_{f_\alpha} \otimes_B B_{g_\beta}.$$

Then A_{ij} a finitely generated B_i -algebra and A_{f_α} a cyclic A_{ij} -algebra gives that A_{f_α} is a finitely generated B_i -algebra. That is, there exists a surjective map

$$B_i[x_1, \dots, x_{n_\alpha}] \rightarrow A_{f_\alpha}.$$

Right exactness of $- \otimes_B B_{g_\beta}$ gives us a surjective map

$$B_{g_\beta}[x_1, \dots, x_{n_\alpha}] \rightarrow A_{f_{\alpha\beta}},$$

whence $A_{f_{\alpha\beta}}$ is a finitely generated B_{g_β} -algebra.

Combining these covers of the $V_i \cap V$ and $U_{ij} \cap U$ gives us a cover $\{D(g_\beta)\}_\beta$ of $\text{Spec } B$, and covers $\{D(f_{\alpha\beta})\}_\alpha$ of each $f^{-1}(D(g_\beta)) = D(\varphi(g_\beta))$, where $\varphi_{g_\beta} : B_{g_\beta} \rightarrow A_{f_{\alpha\beta}}$ is of finite type. Since $\text{Spec } B$ and $D(\varphi(g_\beta))$ are affine, we may assume each of these covers is finite, completing our result. $\square$

Thus, it suffices to show the following lemma.

Lemma. *Let $\varphi : B \rightarrow A$ be a morphism of rings such that $\varphi_i : B \rightarrow A_{f_i}$ is of finite type for each $1 \leq i \leq r$, where $(f_1, \dots, f_r) = (1)$. Then φ is of finite type.*

Proof. Take $\{a_{ij} = a'_{ij}/f_i^{e_{ij}}\}_{j=1}^{r_i}$ to be a set of generators for A_{f_i} as a B -algebra. Without loss of generality, $r_i = r$ for all $1 \leq i \leq r$. Define

$$A' := \varphi(B)[a'_{ij} : 1 \leq i, j \leq r],$$

noting that

$$A'_{f_i} = \varphi(B)[f_i^{-1}, a'_{ij} : 1 \leq j \leq r] = A_{f_i}.$$

Now, consider the quotient A/A' . It suffices to show that $A/A' = 0$. For any prime ideal $\mathfrak{p} \in \text{Spec } A$, we can choose i such that $f_i \notin \mathfrak{p}$. Then

$$\begin{aligned} (A/A')_{\mathfrak{p}} &= \frac{A_{\mathfrak{p}}}{A'_{\mathfrak{p}}} \\ &= \frac{(A_{f_i})_{\mathfrak{p}}}{(A'_{f_i})_{\mathfrak{p}}} \\ &= \left(\frac{A_{f_i}}{A'_{f_i}} \right)_{\mathfrak{p}} \\ &= 0_{\mathfrak{p}} \\ &= 0 \end{aligned}$$

gives us that in fact $A/A' = 0$, since being zero is a local condition. We conclude that A is a finitely generated B -algebra, as desired. $\square$

$\square$

3.2. A morphism $f : X \rightarrow Y$ of schemes is *quasi-compact* if there is a cover of Y by open affines V_i such that $f^{-1}(V_i)$ is quasi-compact for each i . Show that f is quasi compact if and only if for *every* open affine subset $V \subset Y$, $f^{-1}(V)$ is quasi-compact.

Proof. One implication is clear. Conversely, suppose there exists a cover $\{V_i = \operatorname{Spec} B_i\}$ of Y by open affines such that $f^{-1}(V_i)$ is quasi-compact for each i , and take an open affine $V = \operatorname{Spec} B \subset Y$. Fix i and write

$$V \cap V_i = \bigcup_j D(g_{ij}),$$

where we range over $D(g_{ij}) \subset V \cap V_i$ which are affine in both V and V_i . Then

$$V = \bigcup_{i,j} D(g_{ij}) = \bigcup_{1 \leq i,j \leq n} D(g_{ij})$$

by quasi-compactness of V_i . A finite union of compact sets is compact, so it suffices to show the following lemma.

Lemma. *Let $f : X \rightarrow Y$ be a map of schemes, and $\operatorname{Spec} A \cong U \subset Y$ an affine open such that $f^{-1}(U)$ is quasi-compact. For any $h \in A$, $f^{-1}(D(h))$ is also quasi-compact.*

Proof. Take a cover $\{U_\alpha\}_\alpha$ of $f^{-1}(D(h))$, and note that

$$f$$

□

□

3.3. (*)

- (a) Show that a morphism $f : X \rightarrow Y$ is of finite type if and only if it is locally of finite type and quasi-compact.
- (b) Conclude from this that f is of finite type if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ can be covered by a finite number of open affines $U_j = \operatorname{Spec} A_j$, where each A_j is a finitely generated B -algebra.
- (c) Show also if f is of finite type, then for *every* open affine subset $U = \operatorname{Spec} A \subset f^{-1}(V)$, A is a finitely generated B -algebra.

3.4. Show that a morphism $f : X \rightarrow Y$ is finite if and only if for *every* open affine subset $V = \operatorname{Spec} B$ of Y , $f^{-1}(V)$ is affine, equal to $\operatorname{Spec} A$, where A is a finite B -module.

Proof. Fix an open affine cover $\{V_i \cong \operatorname{Spec} B_i\}$ of Y such that $f^{-1}(V_i) \cong \operatorname{Spec} A_i$ is affine, and each A_i is a finite B_i -module. □

3.6. Let X be an integral scheme. Show that the local ring $\mathcal{O}_\xi$ of the generic point ξ of X is a field. It is called the *function field* of X , and is denoted by $K(X)$. Show also if $U = \operatorname{Spec} A$ is any open affine subset of X , then $K(X)$ is isomorphic to the quotient field of A .

Proof. Fix $U = \operatorname{Spec} A \subset X$ open affine. Then $\operatorname{Cl}_U(\{\xi\}) = \{\xi\}^- \cap U = U$ implies that ξ is the minimal prime ideal of U . That is, $\xi = (0) \in \operatorname{Spec} A$. Then $K(X) = A_{(0)}$ is the quotient field of A . □